Assignment -1

Unit -1

Example 1: A coin is thrown 3 times .what is the probability that atleast one head is obtained? Ans:7/8
Example 2: Find the probability of getting a numbered card when a card is drawn from the pack of 52 cards. Ans 9/13
Example 3: There are 5 green 7 red balls. Two balls are selected one by one without replacement. Find the probability that first is green and second is red. Ans: 35/132

Example 4: What is the probability of getting a sum of 7 when two dice are thrown? Ans: 1/6
Example 5: 1 card is drawn at random from the pack of 52 cards.
(i) Find the Probability that it is an honor card. Ans: 4/13
(ii) It is a face card. Ans: 3/13
Example 6: Two cards are drawn from the pack of 52 cards. Find the probability that both are diamonds or both are kings. Ans: 0.063
Example 7: Three dice are rolled together. What is the probability as getting at least one '4'? Ans: 91/216

Example 8: A problem is given to three persons P, Q, R whose respective chances of solving it are 2/7, 4/7, 4/9 respectively. What is the probability that the problem is solved?
Ans: (122/147)

Example 9: Find the probability of getting two heads when five coins are tossed. Ans: 5/16
Example 10: What is the probability of getting a sum of 22 or more when four dice are thrown? Ans: 5/432
Example 11: Two dice are thrown together. What is the probability that the number obtained on one of the dice is multiple of number obtained on the other dice? Ans: 11/18
Example 12: From a pack of cards, three cards are drawn at random. Find the probability that each card is from different suit. Ans: 169/425
Example 13: Find the probability that a leap year has 52 Sundays. Ans: 5/7
Example 14: Fifteen people sit around a circular table. What are odds against two particular people sitting together? Ans: 6:1
Example 15: Three bags contain 3 red, 7 black; 8 red, 2 black, and 4 red & 6 black balls respectively. 1 of the bags is selected at random and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the third bag. Ans: 4/15

Example 16: One bag contains 4 white balls and 3 black balls and a second bag contains 3 white balls and 5 black balls, one ball is drawn from the first bag and placed unseen in the second bag, what is the probability that a ball now drawn from the second bag is black ? Ans.= 38/63

Example 17: Three news papers A, B and C are published in a certain city. A survey shows that for the adult population 20% read A, 16% read B and 14% read C, 8% read both A and B, 5% read both A and C, 4% read both B and C and 2% read all three. If an adult is chosen at, random, find the probability that (i) He reads none of the papers. Ans.= 65% (ii) He reads only one of these papers. Ans.= 22%